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1st Sem / Electrical

Subject :Principles of Electrical Engineering

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The SI unit of electrical current is:

a) Volt b) Ampere

c) Ohm d) Coulomb

Q.2 The property of a coil by which it opposes any change in current is called:

a) Capacitance b) Resistance

c) Inductance d) Conductance

Q.3 The energy stored in a capacitor is expressed as:

a) $1/2CV^2$ b) IR

c) LI^2 d) VI

Q.4 Which of the following is unit of inductance?

a) Ohm b) Henery

c) Faraday d) none

Q.5 ________ in magnetic circuit is equivalent to resistance in electric circuit

a) Flux b) Reluctance

c) mmf d) Magnetic Field

Q.6 Which electrolyte is used in Lead-Acid cells?

a) Diluted H_2SO_4 b) NaOH

c) Concentrated H_2SO_4 d) NaCl

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write the definition of resistance.

Q.8 State True or False: "An ideal voltage source has zero internal resistance."

Q.9 Name two uses of capacitors in electric circuits.

Q.10 Draw the symbol for a cell and a battery.

Q.11 The instrument used for measuring electric current is called________

Q.12 Write the formula for Ohm's Law.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Draw a circuit of three resistors in series and explain the total resistance formula.

Q.14 Write care and maintenance of battery.

Q.15 State and explain Fleming's Left Hand Rule.

Q.16 Explain the hysteresis loop.

Q.17 Give principle of self induction.

Q.18 Define magnetic flux and magnetic flux density.

Q.19 Calculate the current through a 10Ω resistor when a voltage of 20V is applied.

Q.20 Explain delta to star conversion.

Q.21 Write a note on eddy current.

Q.22 List down the similarities between magnetic circuit and electric circuit.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the construction, principle and working of lead acid cell.

Q.24 a) Drive an expression for the force between two parallel current carrying conductors.

b) Drive the expression for energy stored in an inductor.

Q.25 Explain the concept of voltage source, current source, connections and their conversion.